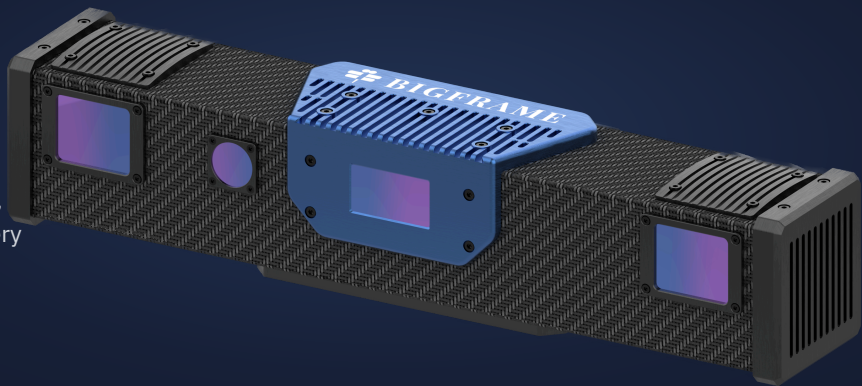


3D cameras for pallets, vehicles and containers

Large scenes at 1.2 to 3.5 m from a bar above the dock, the cell or the conveyor; a color image registered to every point.

PRELIMINARY · VISION 2026



2.56 × 2.06 m

field of view at 2.5 m

1.30 × 1.05 m at 1.2 m; 3.60 × 2.91 m at 3.5 m.

1.2–3.5 m

working distance

2.3 m measuring depth from one fixed mount; 450 mm bar, 3.5 kg, above the scene.

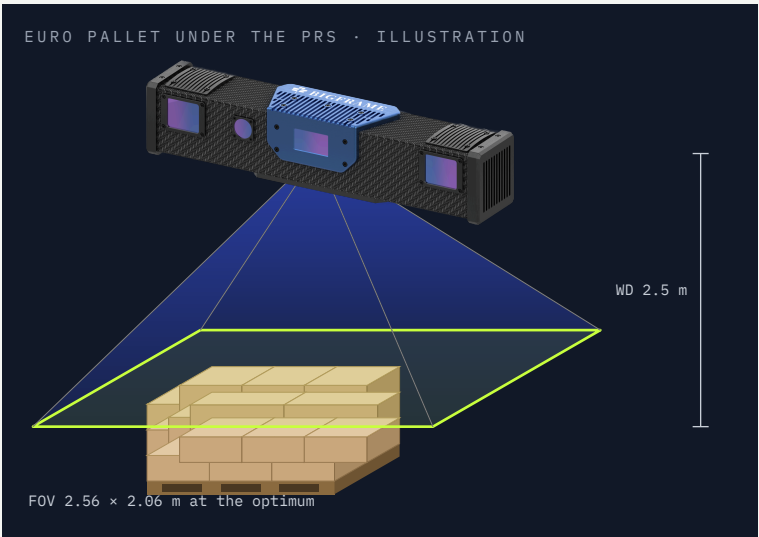
300 μm

Z repeatability ¹

With 1400 μm point spacing at 2.5 m; two 5 MP cameras, RGB image, 1 fps.

The PRS covers the largest scenes in the range — a pallet, a vehicle, a container, a stack on a dock — from one capture, with the color image the operator needs to see what was measured.

The PRS in use



At 2.5 m the field of view of 2.56 × 2.06 m covers a Euro pallet with margin; layer height and carton positions come from one capture.

Which large-scene camera?

Choose the VR-2300B where the finest point spacing at 2 to 4 m matters and no color image is needed; the VRD-1200B for a camera on the arm; the PRS where a scene of 2 m or more must be covered from a fixed mount with a color image.

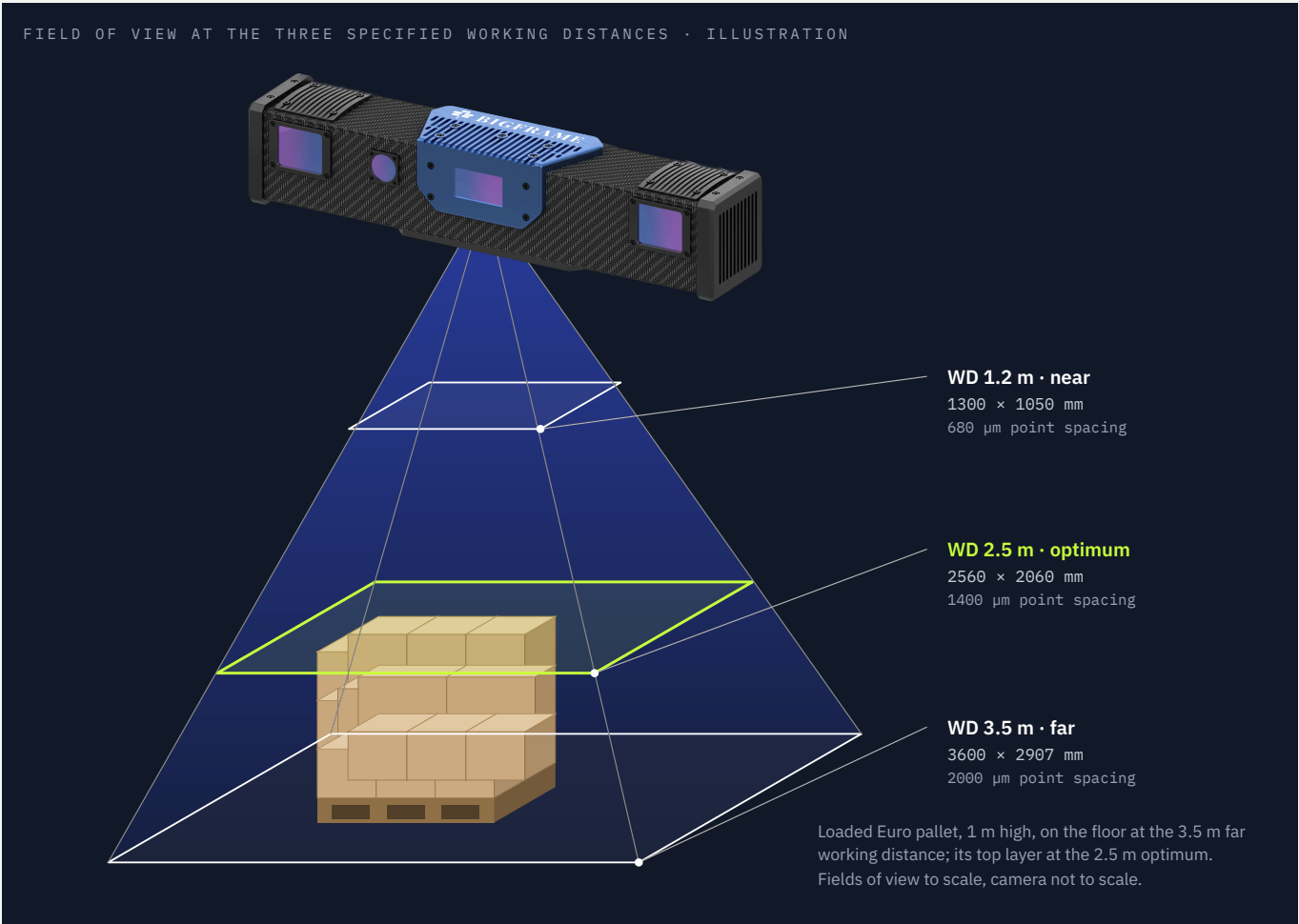
The PRS-2500B is specified at three working distances from 1.2 to 3.5 m (table on page 14); at 3.5 m the field of view of 3.60 × 2.91 m covers a container door with margin; at 1.2 m it is 1.30 × 1.05 m.

Large-scene cameras compared

Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	XY point spacing (μm)	Z repeatability 2σ (μm)	Color image	Mounting
VR-1300B	865 × 645	900 / 1500 / 2200	420	60	No	fixed, above the scene
VR-2300B	1500 × 1115	1900 / 2600 / 3800	730	100	No	fixed, above the scene
VRD-1200B	1166 × 731	900 / 1200 / 1600	480	24	Yes	robot arm
PRS-2500B ¹	2560 × 2060	1200 / 2500 / 3500	1400	300	Yes	fixed, above the scene

¹ PRS: preliminary specification; Z repeatability 300 μm, method to be stated at launch.

PRS series: field of view and dimensions

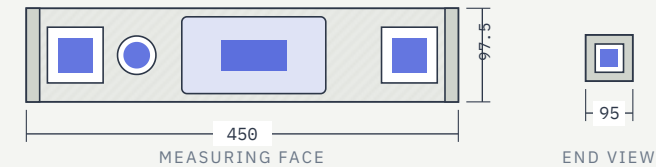


Field of view by working distance ¹

WD (mm)	Field of view (mm)	XY point spacing (µm)
1200 near	1300 × 1050	680
2500 opt.	2560 × 2060	1400
3500 far	3600 × 2907	2000

¹ Preliminary specification; final data sheet at launch. Z repeatability 300 µm, method to be stated at launch. Measuring depth 2300 mm, near to far.

Dimensions and mounting



Housing L × W × H 450 × 97.5 × 95 mm
Weight 3.5 kg
Stereo baseline approx. 350 mm
Interface Gigabit Ethernet; 24 V DC, up to 4 A
Mounting Bracket kit; details at launch

Preliminary specification; dimensions over the housing, connectors excluded. Dimension drawings and 3D CAD data on request.

Your pallet or vehicle,
measured before you buy.

Send a photograph of the scene and the measurement you need to info@bigframe.com — we come back with the model and the working distance.

All models

The figures that decide the choice, in catalog order. Full specifications on pages 18 to 20; definitions and the repeatability method on page 22.

Model	Field of view at optimum WD (mm)	Working distance near–far (mm)	Optimum WD (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹	Points per frame	Interface	L × W × H (mm)	Weight (kg)	Page
● TRS series — Four-projection structured-light 3D cameras											
TRS-010B	9.8 × 9.8	62–63	62.5	1	1.9	0.06	26.2 million	CoaXPress	230 × 230 × 250	8.85	5
TRS-025B	25.5 × 25.5	101.5–106.5	104	5	5	0.1	26.2 million	CoaXPress	240 × 240 × 275	9.74	5
TRS-050B ²	51 × 51	on request	131	8	10	0.2	26.2 million	CoaXPress	300 × 300 × 253	11	5
TRS-075B ²	74 × 74	on request	191	8	14	1	26.2 million	CoaXPress	360 × 360 × 350	17	5
● VRH9 series — Area-scan structured-light 3D cameras, fine measurement											
VRH9-020B	24.3 × 12.5	112–114.5	113	2.5	5.8	0.15	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-040B	48.2 × 24.7	243–257	250	14	11	0.23	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-080B	81.5 × 41.5	260–280	270	20	19	0.36	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-120B	126 × 64.5	260–290	275	30	30	0.5	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-160B	159 × 81.5	340–380	360	40	38	0.95	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-200B	205 × 105	330–390	360	60	49	1.3	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-280B	277 × 142	450–550	500	100	66	1.8	9.07 million	GigE	190 × 140 × 60	1.8	7
● VR series — Area-scan structured-light 3D cameras, medium and large scenes											
VR-400B	300 × 220	350–650	500	300	150	2	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-600B	415 × 305	500–900	700	400	200	14	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-1300B	865 × 645	900–2200	1500	1300	420	60	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-2300B	1500 × 1115	1900–3800	2600	1900	730	100	3.2 million	GigE	220 × 120 × 56	1.6	9
● VRD series — Color stereo structured-light 3D cameras, robot guidance											
VRD-600B	580 × 369	400–800	600	400	240	6	5 million	GigE	166 × 90 × 54	0.87	11
VRD-1200B	1166 × 731	900–1600	1200	700	480	24	5 million	GigE	166 × 90 × 54	0.87	11
● PRS series — Wide-baseline stereo 3D cameras for large scenes · preliminary ³											
PRS-2500B ³	2560 × 2060	1200–3500	2500	2300	1400	300	on request	GigE	450 × 97.5 × 95	3.5	13

¹ Regional 2σ on a ceramic reference plate; the largest value found at the near, optimum and far working distance; method on page 22. ² TRS-050B and TRS-075B: near and far working distances on request; measuring depth 8 mm. ³ PRS: preliminary specification; Z repeatability method and points per frame stated at launch.

Selecting a model

Choose a **field of view** that covers the part with margin at the optimum working distance.

Choose a **point spacing** of one tenth of the smallest feature or finer.

Choose a **Z repeatability** of one tenth of the tolerance or better; a gauge study on the part decides.

Which model? Ask us.

Send a photograph of the part, the feature and its tolerance to info@bigframe.com. We come back with the model and the working distance, and say what the camera will and will not resolve.

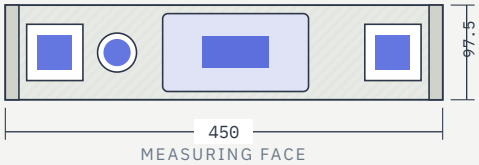
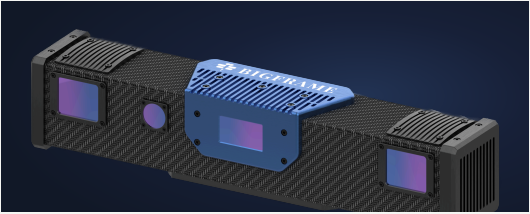


PRS series: technical data

PRS series

NEW

Model	PRS-2500B
Working distance near / optimum / far (mm)	1200 / 2500 / 3500
Field of view at 1200 / 2500 / 3500 mm, W × H (mm)	1300 × 1050 / 2560 × 2060 / 3600 × 2907
Measuring depth, Z range (mm)	2300
XY point spacing at 1200 / 2500 / 3500 mm (μm)	680 / 1400 / 2000
Z repeatability (μm) ¹	300
Camera resolution	2448 × 2048 pixels, 5 MP, 2 cameras
Frame rate	1 fps at full resolution
Optical configuration	Two cameras, stereo baseline approx. 350 mm; laser pattern projector; RGB camera
Light source	Red laser, 638 nm
Laser class	Stated at launch
Color image output	RGB image registered to the point cloud
Data interface	Gigabit Ethernet
Power	24 V DC, up to 4 A
Operating temperature	0 to 45 °C
Dimensions L × W × H (mm)	450 × 97.5 × 95
Weight (kg)	3.5
Software	TridiVision



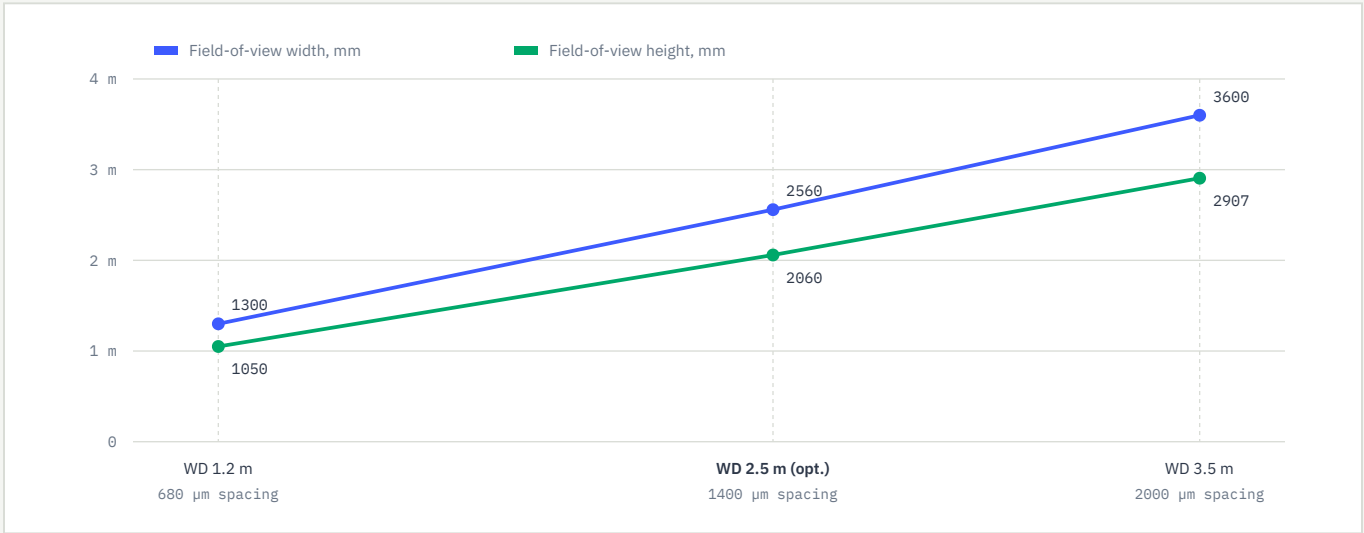
PRS, measuring face; housing 450 × 97.5 × 95 mm, camera windows at both ends, RGB camera and laser projector module between them.

By working distance

WD (mm)	Field of view (mm)	XY point spacing (μm)
1200 near	1300 × 1050	680
2500 opt.	2560 × 2060	1400
3500 far	3600 × 2907	2000

Preliminary specification; final data sheet at launch. ¹ Method to be stated at launch. Dimensions over the housing, connectors excluded.

Field of view and point spacing against working distance



Field-of-view width and height at the three specified working distances, with the XY point spacing under each. At 2.5 m the field of view covers a Euro pallet with margin; at 3.5 m it covers a container door.